BLM3620 Digital Signal Processing

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Short time furier transform
↳ window size

overlapping
çakışma oranı
0,25
L: 500
0-500
250-750
500-1000

Slayttan
hamming pencere
kodu var
onu uygulayacağız

3. soru eğer işaret periyodik değilse furier dönüşümünü alınca sonuç kötü olabiliyor. Merkezde daha güçlü değeri olan pencere kullanırsak, yan taraftaki hatalar minimuma indiriliyor

Lecture #13 – z - Transform

- Introduce the z-Trasnform
- Examples
- Transform Example
- MATLAB Applications

Course Materials

Important Materials:

- James H. McClellan, R. W. Schafer, M. A. Yoder, *DSP First Second Edition*, Pearson, 2015.

- Lizhe Tan, Jean Jiang, *Digital Signal Processing: Fundamentals and Applications*, Third Edition, Academic Press, 2019.

Auxilary Materials:

- Prof. Sarp Ertürk, *Sayısal İşaret İşleme*, Birsen Yayınevi.

- Prof. Nizamettin Aydin, DSP Lecture Notes.

- J. G. Proakis, D. K. Manolakis, *Digital Signal Processing Fourth Edition*, Peason, 2014.

- J. K. Perin, *Digital Signal Processing*, *Lecture Notes,* Standford University, 2018.

Syllabus

Week	Lectures
1	Introduction to DSP and MATLAB
2	Sinuzoids and Complex Exponentials
3	Spectrum Representation
4	Sampling and Aliasing
5	Discrete Time Signal Properties and Convolution
6	Convolution and FIR Filters
7	Frequency Response of FIR Filters
8	Midterm Exam
9	Discrete Time Fourier Transform and Properties
10	Discrete Fourier Transform and Properties
11	Fast Fourier Transform and Windowing
12	-
13	z- Transforms
14	FIR Filter Design and Applications

For more details -> Bologna page: http://www.bologna.yildiz.edu.tr/index.php?r=course/view&id=5730&aid=3

Review & Recall

- The FFT is simply an algorithm for efficiently calculating the DFT
- Computational efficiency of an N-Point FFT:
  - DFT: N^2 Complex Multiplications
  - FFT: $(N/2)\log_2(N)$ Complex Multiplications

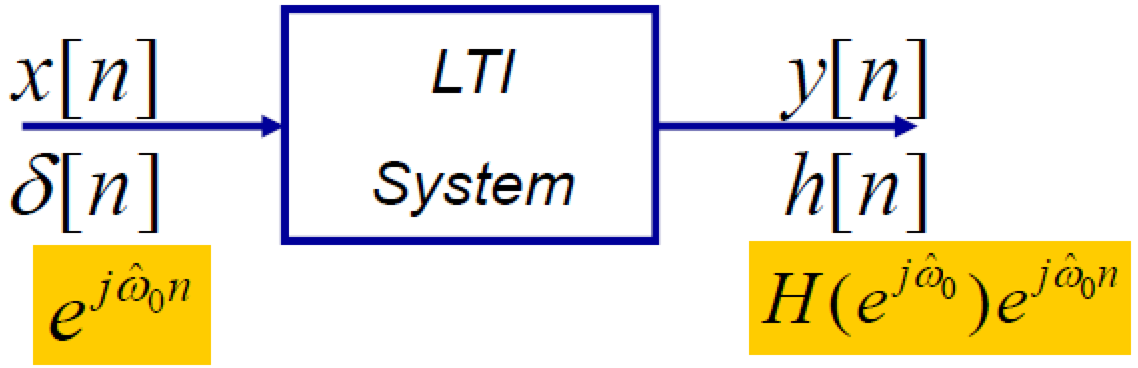
fft'nin hız avantajı

N	DFT Multiplications	FFT Multiplications	FFT Efficiency
256	65,536	1,024	64 : 1
512	262,144	2,304	114 : 1
1,024	1,048,576	5,120	205 : 1
2,048	4,194,304	11,264	372 : 1
4,096	16,777,216	24,576	683 : 1

Discrete Fourier Transform (DFT)

$$X[k]=\sum_{n=0}^{N-1}x[n]e^{-j(2\pi/N)kn}$$

Recap: Frequency Response $H(e^{j\hat{\omega}})$



$$H(e^{j\hat{\omega}}) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\hat{\omega}n}$$

$$\text{Periodic}: H(e^{j(\hat{\omega}+2\pi)}) = H(e^{j\hat{\omega}})$$

$$y[n] = A \cdot \left|H(e^{j\hat{\omega}_0})\right| \cos(\hat{\omega}_0 n + \varphi + \angle H(e^{j\hat{\omega}_0}))$$

Transfer Function/System Function $H(z)$

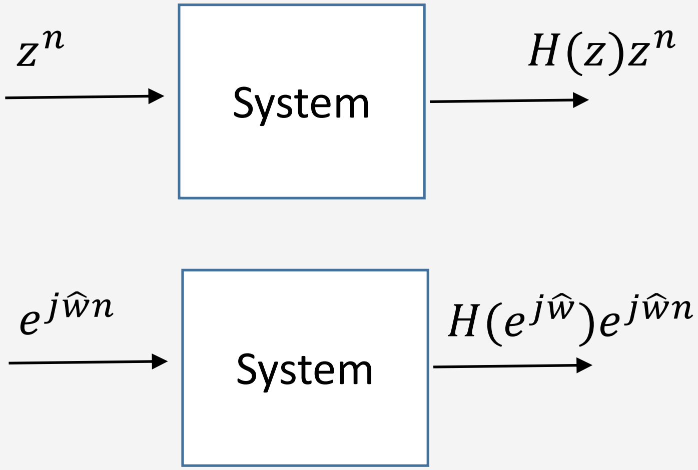
$e^{j\hat{\omega}} = z$
Burada $r = 1$

$X(z) = \sum_{n=-\infty}^{\infty} x[n] . z^{-n}$

Given impulse response of an LTI system h[n], the transfer function can be found as:

$r = -\infty \dots \infty$ için de bu dönüşüm kullanılır.

$$H[z] = \sum_{k=-\infty}^{\infty} h[k]\, z^{-k}$$



Z-transform is a Generalized version of Discrete Time Fourier Transform ($z = re^{j\hat{w}}$)

Z transform can be computed if FT exists or not.

Easy to compute the system output.

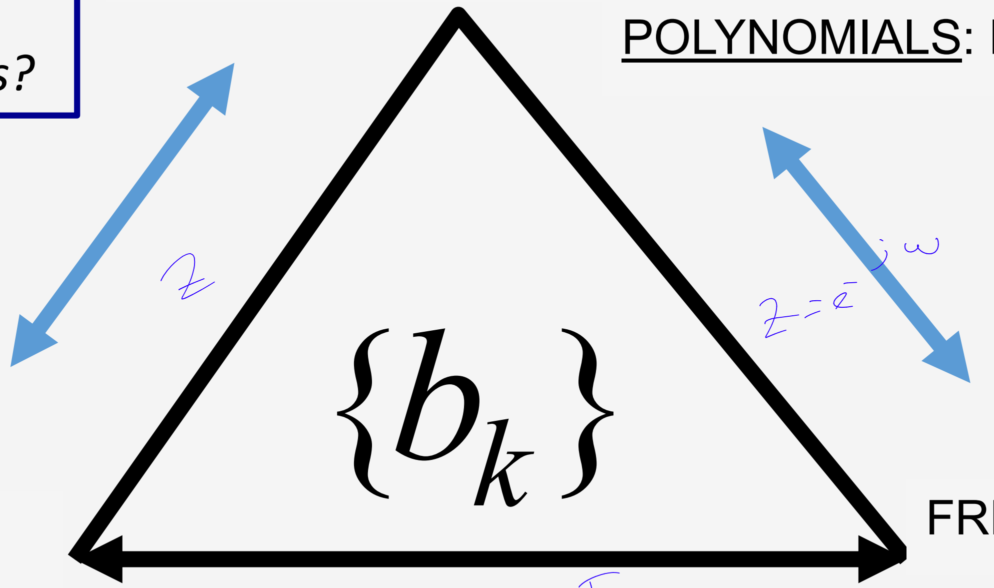
Mostly used with block diagrams to illustrate the system.

Three Domains

Algebra Domain:
Best of both worlds?

Z-TRANSFORM-DOMAIN

POLYNOMIALS: H(z)



TIME-DOMAIN

FREQ-DOMAIN

DTFT

konvolüsyon

$$y[n] = \sum_{k=0}^{M} b_k x[n-k]$$

$$H(e^{j\hat{\omega}}) = \sum_{k=0}^{M} b_k e^{-j\hat{\omega} k}$$

Z-Transform DEFINITION

- POLYNOMIAL Representation of LTI SYSTEM:
- EXAMPLE:

$$H(z) = \sum_{n} h[n] z^{-n}$$

APPLIES to any SIGNAL

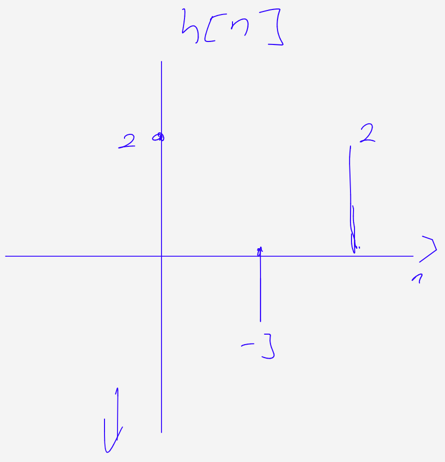
$$\{h[n]\} = \{2, 0, -3, 0, 2\}$$

$$H(z) = 2z^{-0} + 0z^{-1} - 3z^{-2} + 0z^{-3} + 2z^{-4}$$

$$= 2 - 3z^{-2} + 2z^{-4}$$

$$= 2 - 3(z^{-1})^2 + 2(z^{-1})^4$$

POLYNOMIAL in z^{-1}



$h[n] = 2\delta[n] - 3\delta[n-2] + 2\delta[n-4]$

Z-Transform Property: Delay Property

$y[n] = x[n] + 4{,}5 \cdot y[n-1] - 2 \cdot y[n-2]$

$Y(z) = X(z) + 4{,}5 \cdot z^{-1} Y(z) - 2 \cdot z^{-2} \cdot Y(z)$

$Y(z) = H(z) \cdot X(z)$ $\quad H(z) = \frac{Y(z)}{X(z)} = \frac{1}{1 - 4{,}5 \cdot z^{-1} + 2 \cdot z^{-2}}$

A delay of one sample multiplies the z-transform by z^{-1}.

$$x[n-1] \iff z^{-1}X(z)$$

Time delay of n_0 samples multiplies the z-transform by z^{-n_0}

$$x[n-n_0] \iff z^{-n_0}X(z)$$

Example:

Find z-transform of shifted impulse $x[n] = 3\delta[n-5]$.

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]\, z^{-n}$$

z dönüşümü verilmiş işaretin fourier transformunu sorar

$$X(z) = \sum_{n=-\infty}^{\infty} 3\delta[n-5]\, z^{-n} = 3\delta[0] z^{-5} = 3z^{-5}$$

Fourier transform?

$$X\left(e^{j\Omega}\right) = X[z]\Big|_{z=e^{jw}} = 3e^{-j5w}$$

Transfer Fonksiyonu

$y(z) - z^{-1} y(z) = x(z) + 2 \cdot z^{-1} \cdot x(z)$

$y[n]-y[n-1]=x[n]+2x[n-1]$ sisteminin transfer fonksiyonunu bulunuz.

$\frac{1+2 \cdot z^{-1}}{1-z^{-1}}$

Fark denkleminde her iki tarafın z-dönüşümü alındığında

$$Y(z)-z^{-1}Y(z)=X(z)+2z^{-1}X(z)$$

elde edilmektedir. Sistemin transfer fonksiyonu

$$H(z)=\frac{Y(z)}{X(z)}=\frac{1+2z^{-1}}{1-z^{-1}}$$

olarak bulunmaktadır.

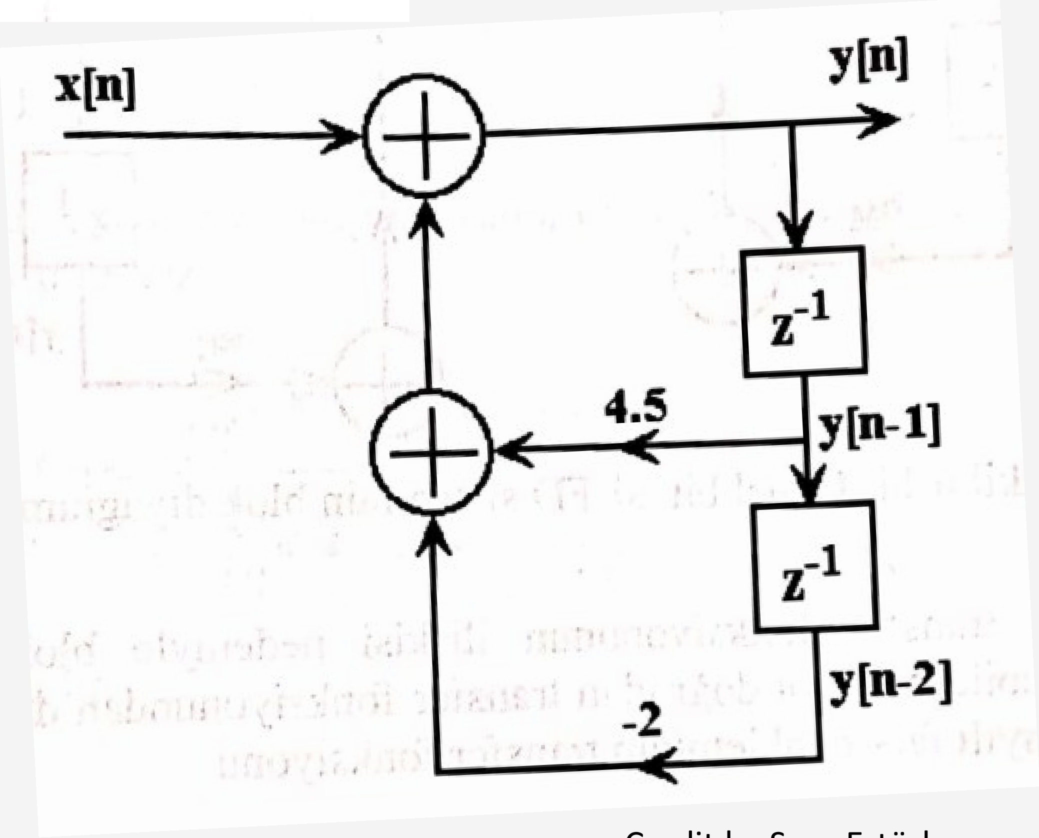
Credit by Sarp Ertürk

Blok Diyagram Gösterimi

$y[n]-4.5y[n-1]+2y[n-2]=x[n]$ şeklinde tanımlanmış sistemin blok diyagramını çizelim.

$$y[n]=x[n]+4.5y[n-1]-2y[n-2]$$

$$H(z)=\frac{1}{1-4.5z^{-1}+2z^{-2}}$$



Credit by Sarp Ertürk

- ANY SIGNAL has a z-Transform:

$$H(z) = \sum_{n} h[n] z^{-n}$$

$$X(z) = \sum_{n} x[n] z^{-n}$$

Example 7.1

n	$n < -1$	-1	0	1	2	3	4	5	$n > 5$
$x[n]$	0	0	2	4	6	4	2	0	0

$$X(z) = ?$$

$$X(z) = 2 + 4z^{-1} + 6z^{-2} + 4z^{-3} + 2z^{-4}$$

Example 9.2

$$X(z) = 1 - 2z^{-1} + 3z^{-3} - z^{-5}$$

ters dönüşüm

EXPONENT GIVES TIME LOCATION

$$x[n] = \begin{cases} 0 & n < 0 \\ 1 & n = 0 \\ -2 & n = 1 \\ 0 & n = 2 \\ 3 & n = 3 \\ 0 & n = 4 \\ -1 & n = 5 \\ 0 & n > 5 \end{cases}$$

$$x[n] = ?$$

$$x[n] = \delta[n] - 2\delta[n-1] + 3\delta[n-3] - \delta[n-5]$$

Z-Transform of FIR Filter

- CALLED the **SYSTEM FUNCTION**
  - because h[n] is same as $\{b_k\}$

SYSTEM FUNCTION

$$H(z) = \sum_{k=0}^{M} b_k z^{-k} = \sum_{k=0}^{M} h[k] z^{-k}$$

$$y[n] = \sum_{k=0}^{M} b_k x[n-k] = \sum_{k=0}^{M} h[k] x[n-k]$$

CONVOLUTION

FIR DIFFERENCE EQUATION

Z-Transform of FIR Filter

- Get H(z) DIRECTLY from the $\{b_k\}$
- Example 7.3 in the book:

$$y[n] = 6x[n] - 5x[n-1] + x[n-2]$$

$$\{b_k\} = \{6, -5, 1\}$$

$$\boxed{H(z) = \sum b_k z^{-1} = 6 - 5z^{-1} + z^{-2}}$$

GENERAL I/O PROBLEM

- Input is x[n], find y[n]   (for FIR, h[n])
- How to combine X(z) and H(z) ?

Example 7.5

$$x[n] = \delta[n-1] - \delta[n-2] + \delta[n-3] - \delta[n-4]$$

$$\text{and} \quad h[n] = \delta[n] + 2\delta[n-1] + 3\delta[n-2] + 4\delta[n-3]$$

$$X(z) = 0 + 1z^{-1} - 1z^{-2} + 1z^{-3} - 1z^{-4}$$

$$\text{and} \quad H(z) = 1 + 2z^{-1} + 3z^{-2} + 4z^{-3}$$

FIR Filter = CONVOLUTION

x[n], X(z)	0	+1	-1	+1	-1			
h[n], H(z)	1	2	3	4				
	0	+1	-1	+1	-1			
		0	+2	-2	+2	-2		
			0	+3	-3	+3	-3	
				0	+4	-4	+4	-4
y[n], Y(z)	0	+1	+1	+2	+2	-3	+1	-4

$$y[n] = \sum_{k=0}^{M} b_k x[n-k] = \sum_{k=0}^{M} h[k] x[n-k]$$

CONVOLUTION

CONVOLUTION PROPERTY

- PROOF:

$$y[n] = x[n] * h[n] = \sum_{k=0}^{M} h[k]x[n-k]$$

$$Y(z) = \sum_{k=0}^{M} h[k]\left(z^{-k}X(z)\right)$$

$$= \left(\sum_{k=0}^{M} h[k]z^{-k}\right) X(z) = H(z)X(z).$$

MULTIPLY
z-TRANSFORMS

CONVOLUTION EXAMPLE

- **<u>MULTIPLY</u>** the z-TRANSFORMS:

Example 7.5

$$x[n] = \delta[n-1] - \delta[n-2] + \delta[n-3] - \delta[n-4]$$

$$\text{and} \quad h[n] = \delta[n] + 2\delta[n-1] + 3\delta[n-2] + 4\delta[n-3]$$

$$X(z) = 0 + 1z^{-1} - 1z^{-2} + 1z^{-3} - 1z^{-4}$$

$$\text{and} \quad H(z) = 1 + 2z^{-1} + 3z^{-2} + 4z^{-3}$$

$$\boxed{y[n] = h[n] * x[n] \quad \leftrightarrow \quad Y(z) = H(z)X(z)}$$

CONVOLUTION EXAMPLE

- Finite-Length input x[n]
- FIR Filter (L=4)

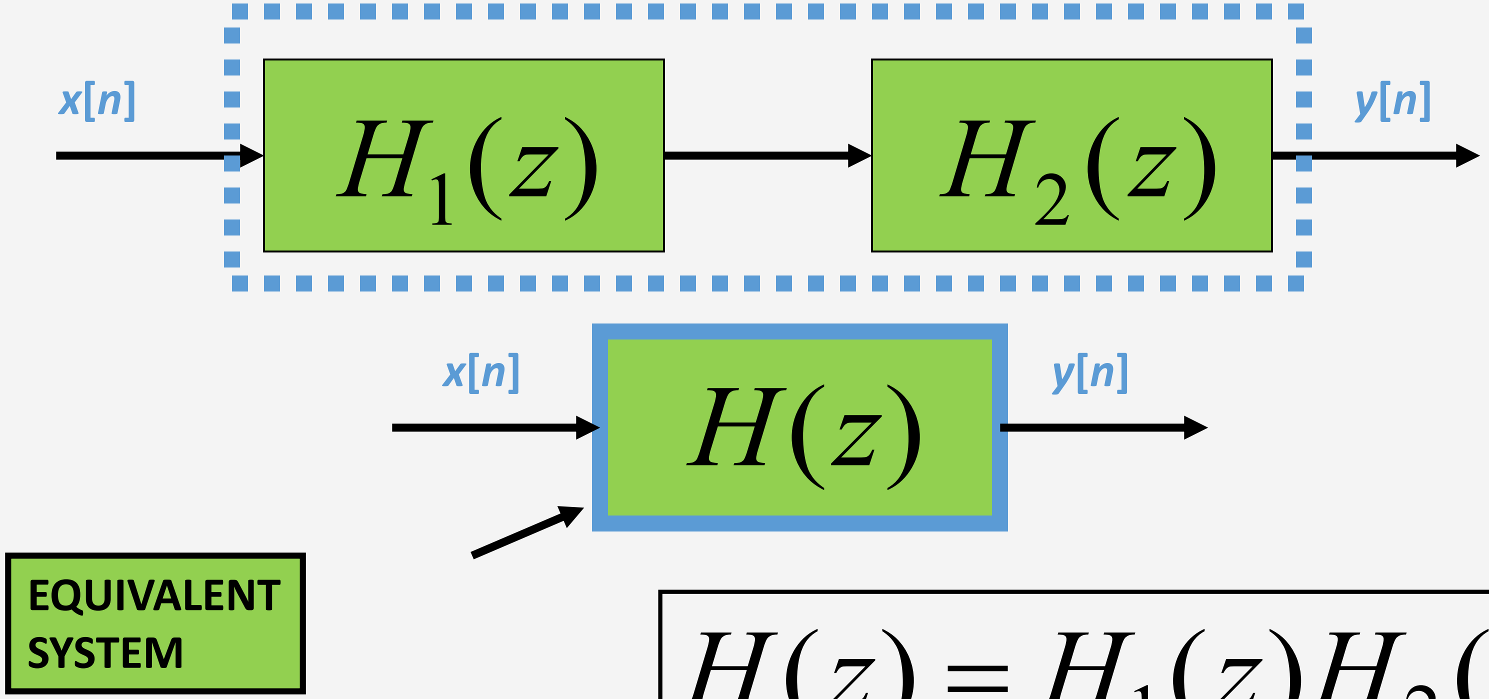
MULTIPLY Z-TRANSFORMS

$$
\begin{aligned}
Y(z) &= H(z)X(z) \\
&= (1 + 2z^{-1} + 3z^{-2} + 4z^{-3})(z^{-1} - z^{-2} + z^{-3} - z^{-4}) \\
&= z^{-1} + (-1 + 2)z^{-2} + (1 - 2 + 3)z^{-3} + (-1 + 2 - 3 + 4)z^{-4} \\
&\quad + (-2 + 3 - 4)z^{-5} + (-3 + 4)z^{-6} + (-4)z^{-7} \\
&= z^{-1} + z^{-2} + 2z^{-3} + 2z^{-4} - 3z^{-5} + z^{-6} - 4z^{-7}
\end{aligned}
$$

$y[n]$ = ?

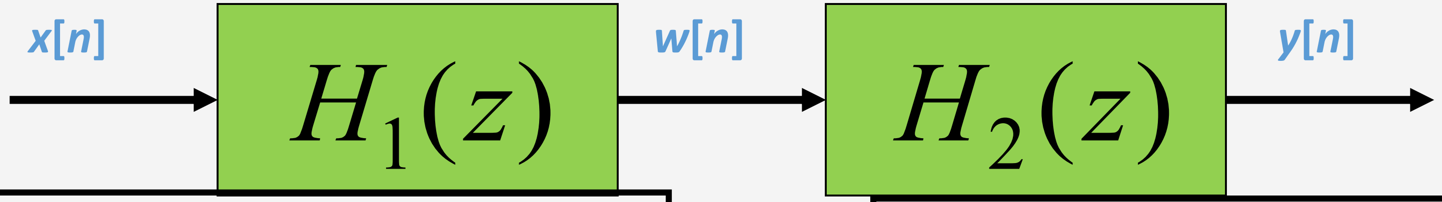
Z-Transform Property: CASCADE EQUIVALENT

- Multiply the System Functions



$$\boxed{H(z) = H_1(z)H_2(z)}$$

CASCADE EXAMPLE

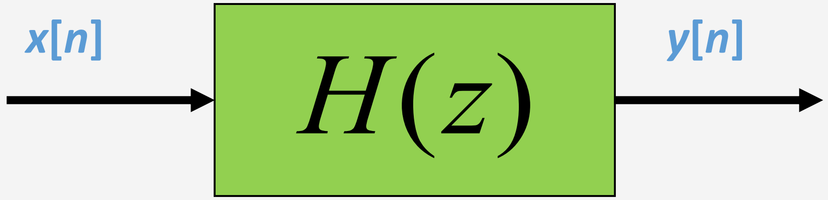


$$\boxed{w[n] = x[n] - x[n-1]}$$

$$\boxed{H_1(z) = 1 - z^{-1}}$$

$$\boxed{y[n] = w[n] + w[n-1]}$$

$$\boxed{H_2(z) = 1 + z^{-1}}$$



$$\boxed{H(z) = (1 - z^{-1})(1 + z^{-1}) = 1 - z^{-2}}$$

$$\boxed{y[n] = x[n] - x[n-2]}$$

$H_1(z) = \frac{w(z)}{x(z)}$

$x(z) - x(z-1) = w(z)$

$1 - z^{-1} = \frac{w(z)}{x(z)}$

Region of Convergence (ROC)

- Give a sequence, the set of values of z for which the z-transform converges, i.e., $|X(z)|<\infty$, is called the region of convergence.

↳ yakınsaklık dönüşümü

z farklı işaretin z dönüşümü aynı olabilir. Bundan dolayı yakınsaklık aralığı belirtmek gerek

$$|X(z)| = \left|\sum_{n=-\infty}^{\infty} x(n) z^{-n}\right| = \sum_{n=-\infty}^{\infty} |x(n)|\,|z|^{-n} < \infty$$

↳ z dönüşümü yapılabilmesi için ↑ <∞ olmalı

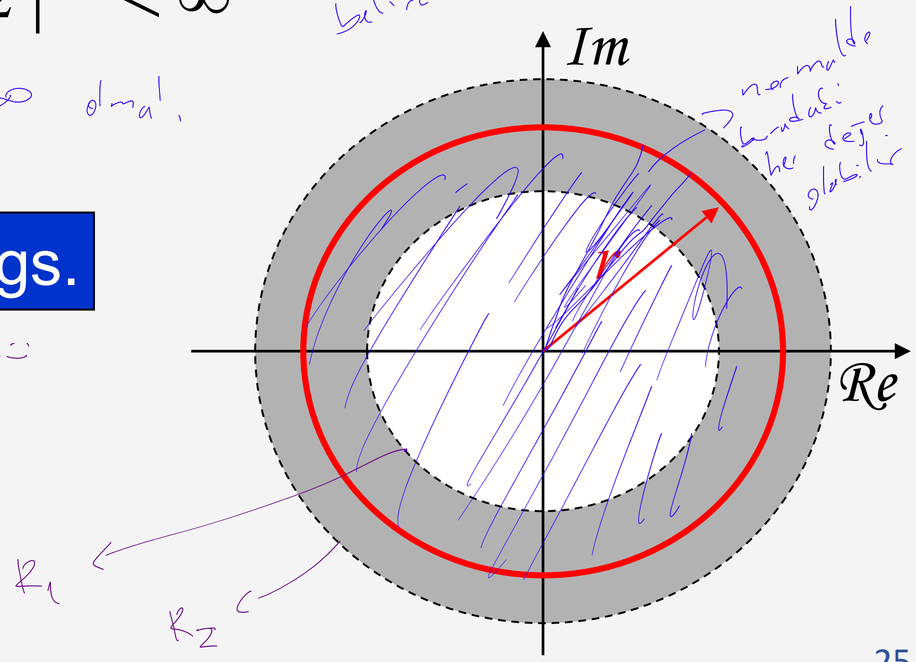
ROC is centered on origin and consists of a set of rings.

Tamamı alan birim çemberi kapsıyorsa fourier dönüşümü alınabilir.

$$ROC = \{z = re^{j\omega} \mid R_{x-} < r < R_{x+}\}$$

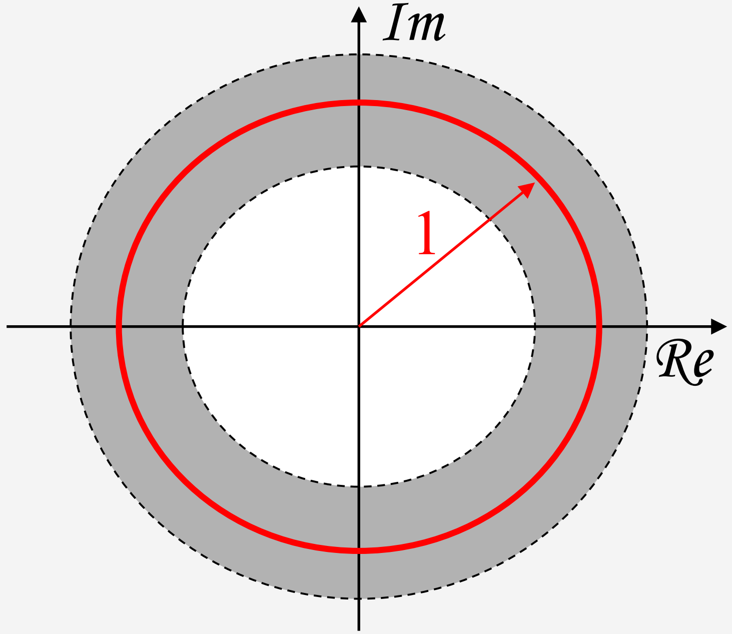
↳ r·cos(ŵ) + j·r·sin(ŵ)

z bu aralıkta tanımlı



Stable Systems

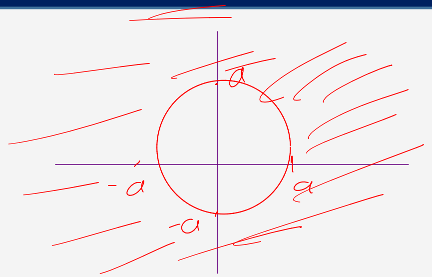
- A stable system requires that its Fourier transform is uniformly convergent.



- Fact: Fourier transform is to evaluate z-transform on a unit circle.
- A stable system requires the ROC of z-transform to include the unit circle. →Stabil sistem birim çemberi kapsamalı, kapsarsa Stable ve FT hesaplanabilir

Example: A right sided Sequence

birim çemberi kapsarsa Stable



a = 0.5 olsa Stable ve Ff tersi alınabilir olur

$$\boxed{x(n) = a^n u(n)}$$

$$X(z) = \sum_{n=-\infty}^{\infty} a^n u(n) z^{-n}$$

$$= \sum_{n=0}^{\infty} a^n z^{-n}$$

$$= \sum_{n=0}^{\infty} (az^{-1})^n$$

For convergence of $X(z)$, we require that

$$\sum_{n=0}^{\infty} |az^{-1}| < \infty \Rightarrow |az^{-1}| < 1$$

$$\Rightarrow |z| > |a|$$

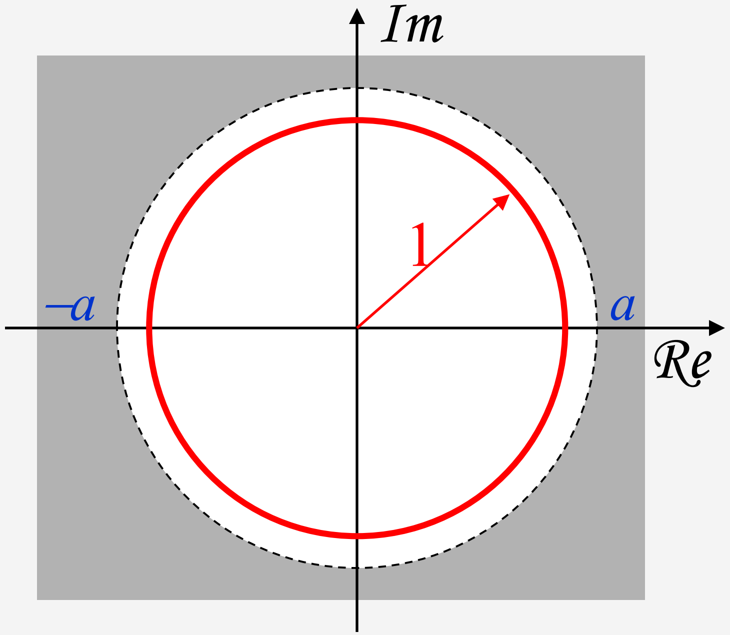
$$X(z) = \sum_{n=0}^{\infty} (az^{-1})^n = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}$$

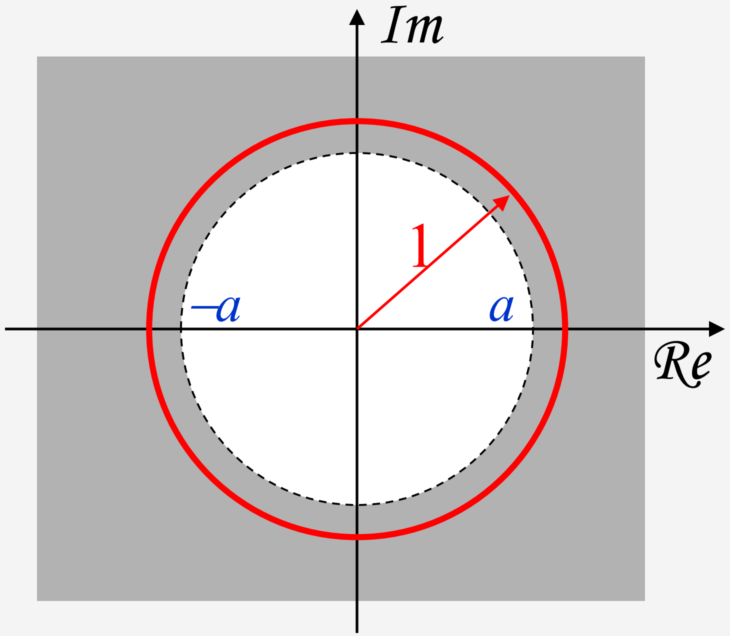
$$|z| > |a|$$

Example: A right sided Sequence ROC for $x(n)=a^n u(n)$

$$X(z) = \frac{z}{z-a}, \qquad |z| > |a|$$

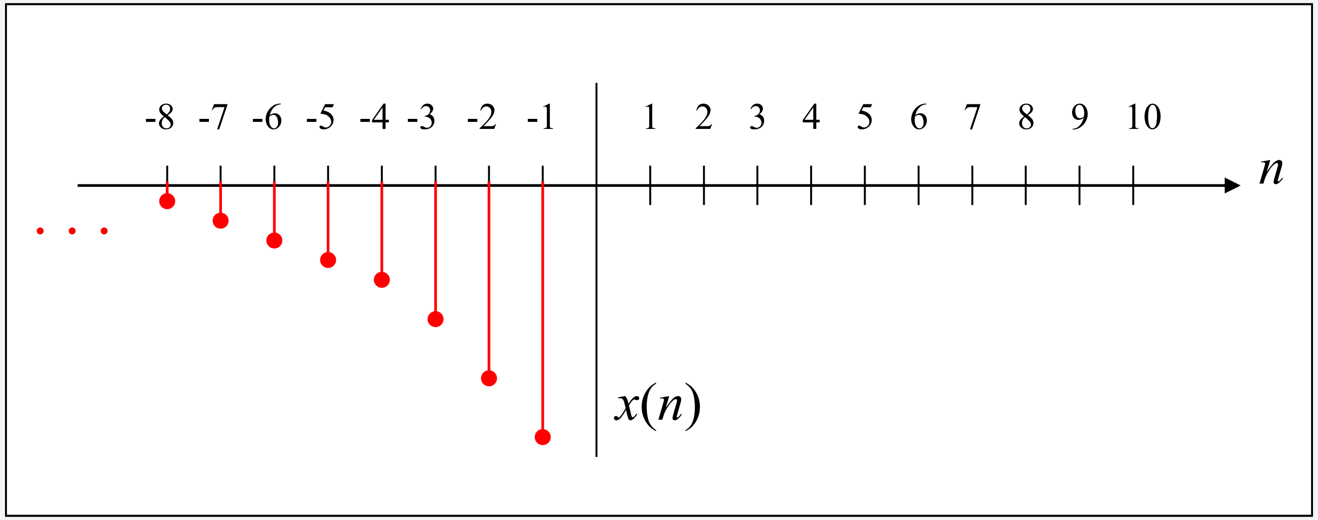
Which one is stable?





Another Example

$$x(n) = -a^n u(-n-1)$$



Example: A left sided Sequence

$$\boxed{x(n) = -a^n u(-n-1)}$$

← Bunun z dönüşümüyle öncekinin dönüşümü eşit

$$X(z) = -\sum_{n=-\infty}^{\infty} a^n u(-n-1) z^{-n}$$

$$= -\sum_{n=-\infty}^{-1} a^n z^{-n}$$

$$= -\sum_{n=1}^{\infty} a^{-n} z^n$$

$$= 1 - \sum_{n=0}^{\infty} a^{-n} z^n$$

For convergence of $X(z)$, we require that

$$\sum_{n=0}^{\infty} |a^{-1} z| < \infty \Rightarrow |a^{-1} z| < 1$$

$$\Rightarrow |z| < |a|$$

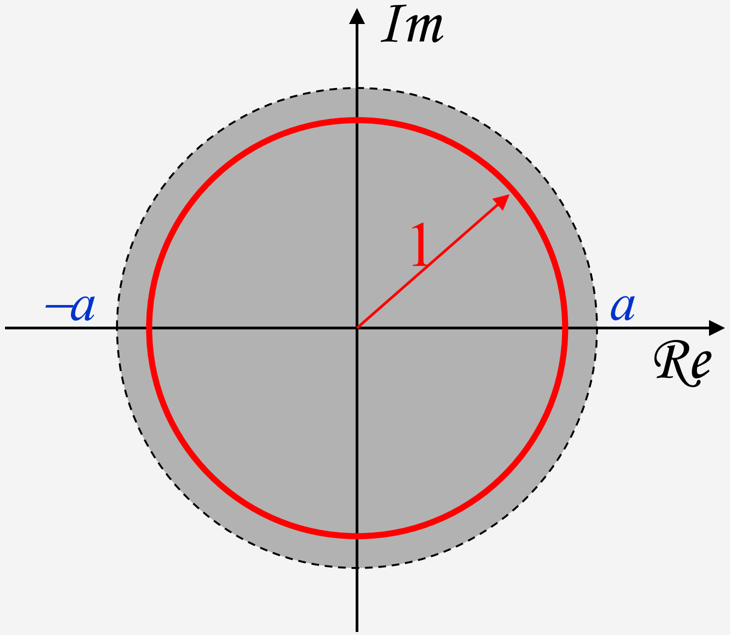
$$X(z) = 1 - \sum_{n=0}^{\infty} (a^{-1} z)^n = 1 - \frac{1}{1 - a^{-1} z} = \frac{z}{z-a}$$

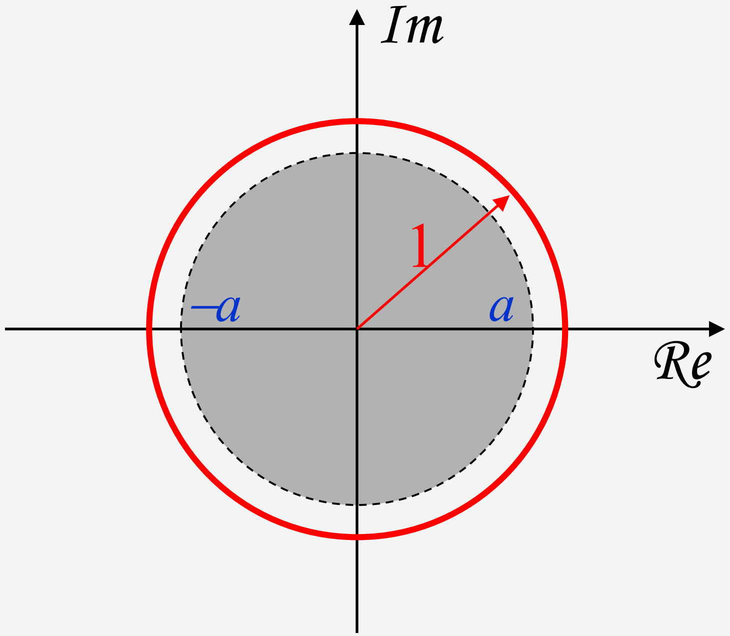
$$|z| < |a|$$

Example: A left sided Sequence ROC for $x(n) = -a^n u(-n-1)$

$$X(z) = \frac{z}{z-a}, \qquad |z| < |a|$$

Which one is stable?





Example: A right sided Sequence

Yakınsaklık bölgesi kuralları

Sağ → (1) x[n] işareti nedensel ise

sol → (2) x[n] işareti antikausalsa

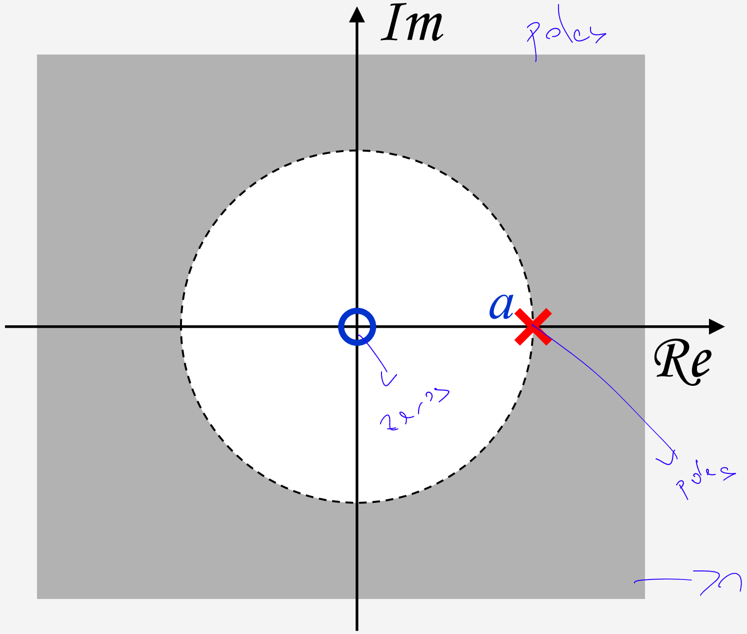
çift → (3) x[n] işareti çift yanlı ise

(1) $a^n u[n]$ → (2) (3)

zeros

$$x(n) = a^n u(n) \Rightarrow X(z) = \frac{z}{z-a}, \quad |z| > |a|$$

poles

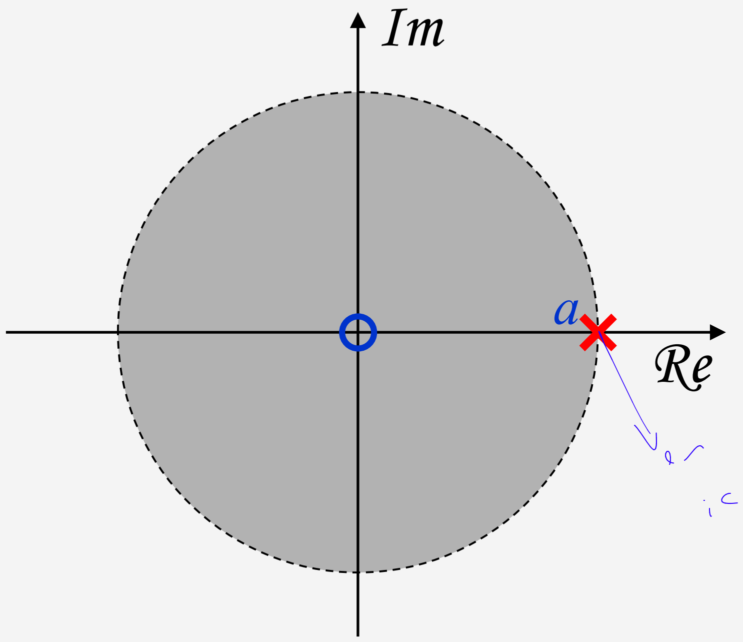


ROC is **bounded by the pole** and is the **exterior of a circle**.

→ nedensel ve kutbu a olduğu için a'dan sonrası taranır en dıştakinden dışa

Example: A left sided Sequence

$$x(n) = -a^n u(-n-1) \Rightarrow X(z) = \frac{z}{z-a}, \qquad |z| < |a|$$



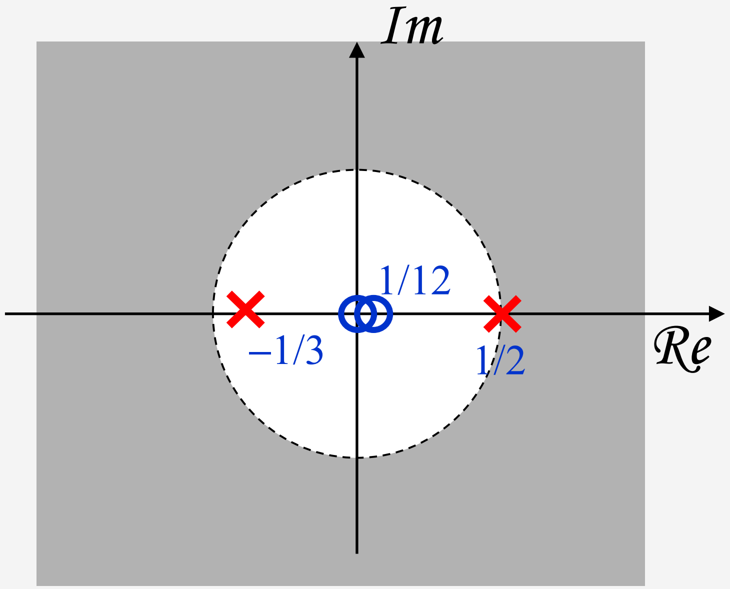
er içtekinden
içi

ROC is bounded by the pole and is the interior of a circle.

Example: Sum of Two Right Sided Sequences

$$x(n) = (\tfrac{1}{2})^n u(n) + (-\tfrac{1}{3})^n u(n)$$

$$X(z) = \frac{z}{z-\frac{1}{2}} + \frac{z}{z+\frac{1}{3}} = \frac{2z(z-\frac{1}{12})}{(z-\frac{1}{2})(z+\frac{1}{3})}$$



ROC is bounded by poles and is the exterior of a circle.

ROC does not include any pole.

Example: A Two Sided Sequence

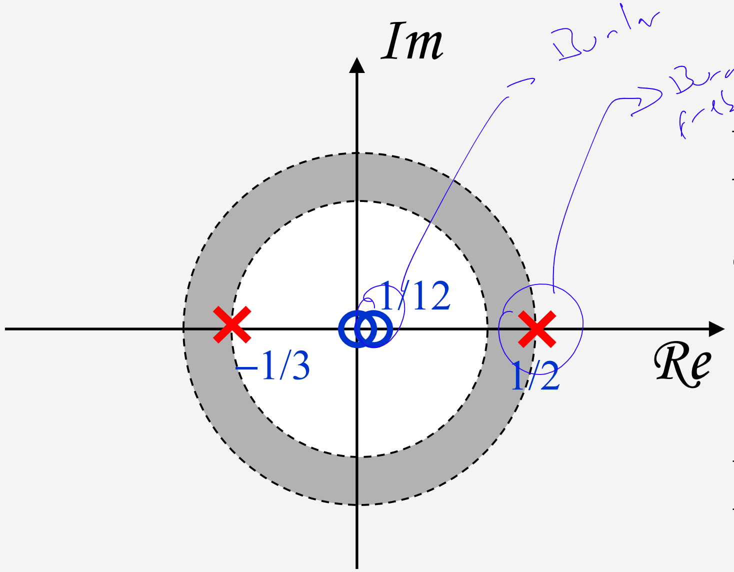
ayrı ayrı z dönüşümü alınabilir

$$x(n) = (-\tfrac{1}{3})^n u(n) - (\tfrac{1}{2})^n u(-n-1)$$

$$\Rightarrow \quad X(z) = \frac{z}{z+\frac{1}{3}} + \frac{z}{z-\frac{1}{2}} = \frac{2z(z-\frac{1}{12})}{(z+\frac{1}{3})(z-\frac{1}{2})}$$

Bunlar da azaltıyor

Buradaki kutup frekansın etkisini artırıyor



ROC is bounded by poles and is a ring.

ROC does not include any pole.

Represent z-transform as a Rational Function

$$X(z) = \frac{P(z)}{Q(z)}$$

where $P(z)$ and $Q(z)$ are polynomials in z.

Zeros: The values of z's such that $X(z) = 0$

Poles: The values of z's such that $X(z) = \infty$

Sıfır / Kutup Gösterimi

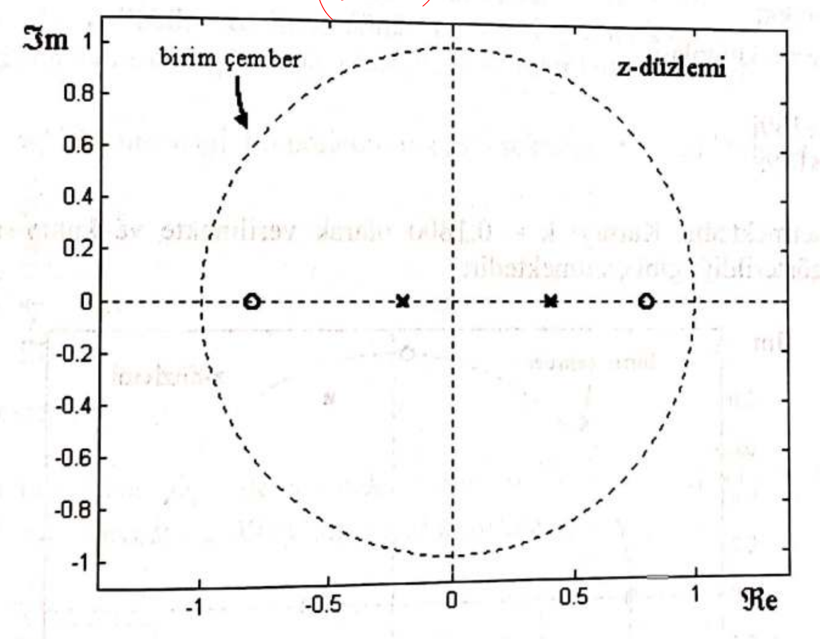
$$X(z)=\frac{1-0.64z^{-2}}{1-0.2z^{-1}-0.08z^{-2}}$$ için kutup-sıfır grafiğini çizelim.

$z^2-0.64 \rightarrow -0.8, +0.8 \rightarrow$ zeros

$z^2-0.2z-0.08 \rightarrow$ poles $\rightarrow 0.4, -0.2$

$(z-0.4)(z+0.2)$

$$X(z)=\frac{\left(1-0.8z^{-1}\right)\left(1+0.8z^{-1}\right)}{\left(1-0.4z^{-1}\right)\left(1+0.2z^{-1}\right)}$$



Credit by Sarp Ertürk

POLES of H(z)

- Find z, where
  - FIR only has poles at z=0

$$H(z) \rightarrow \infty$$

$$H(z) = \frac{z^3 - 2z^2 + 2z \;\; - 1}{z^3}$$

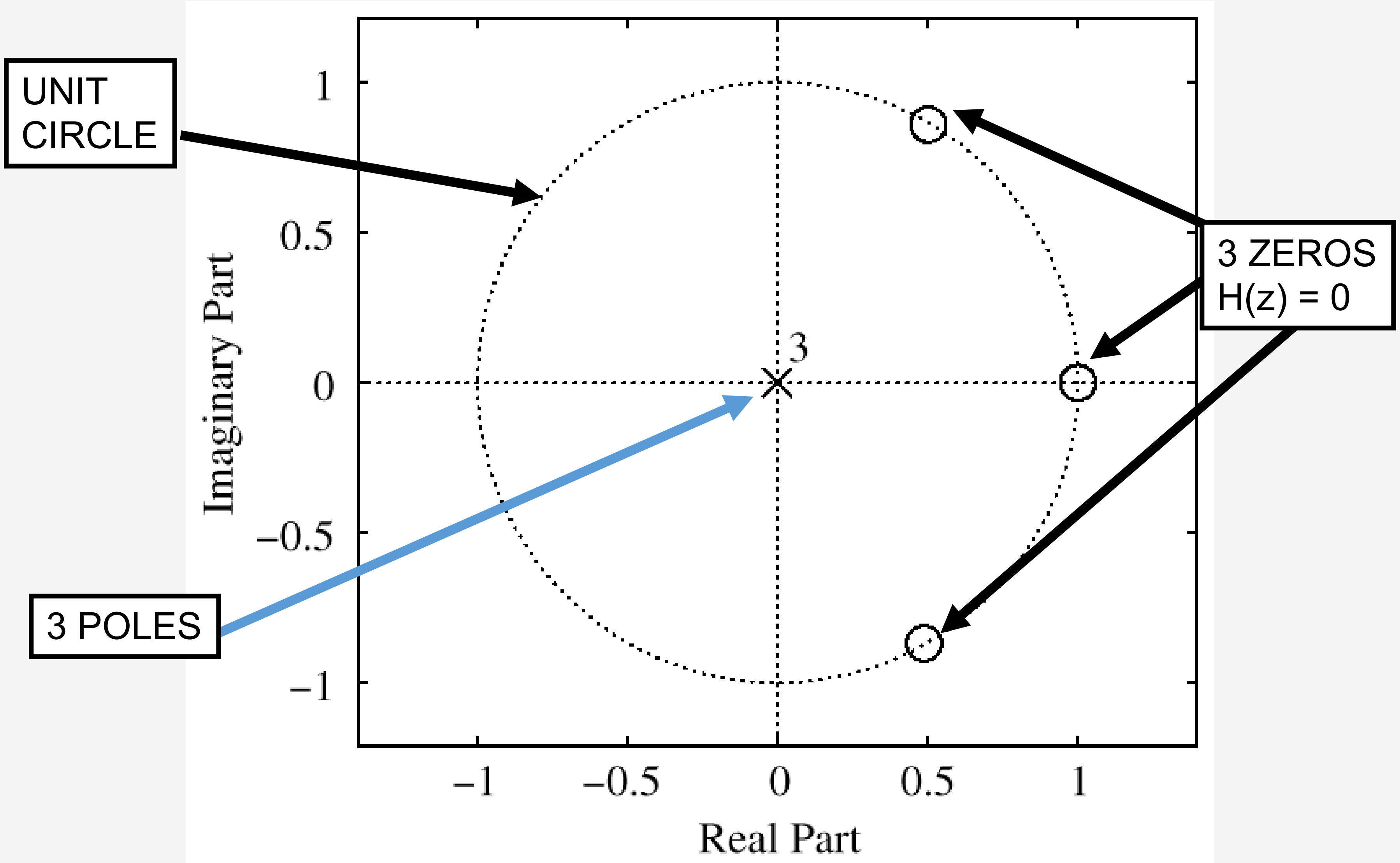
$$\boxed{\text{Three Poles at}: z = 0}$$

$$\boxed{\text{Roots}: z = 1, \frac{1}{2} \pm j\frac{\sqrt{3}}{2}}$$

$$\boxed{e^{\pm j\pi/3}}$$

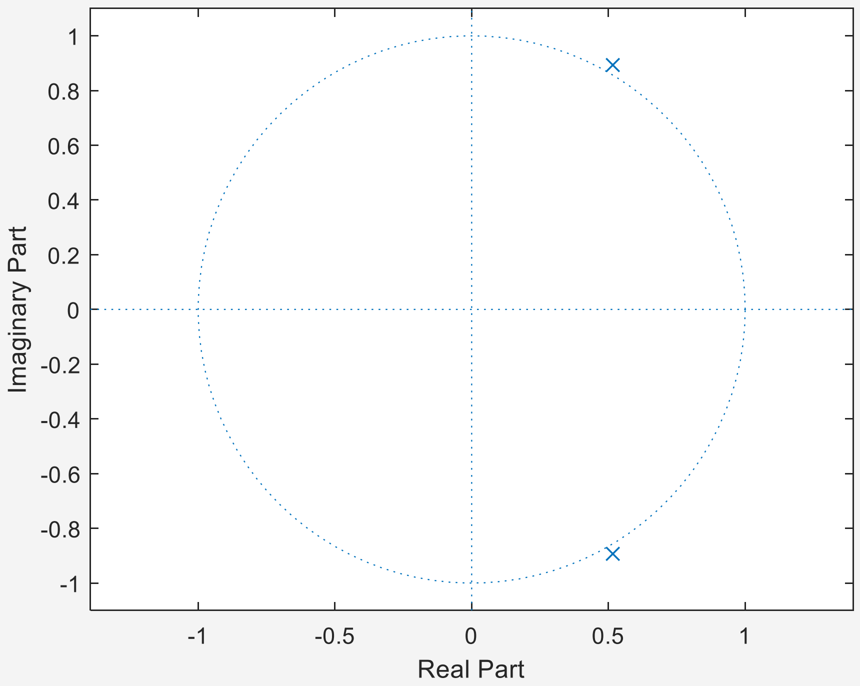
Bu tip zor
kök çıkmaz
hoca dedi

PLOT ZEROS in z-DOMAIN



MATLAB Code

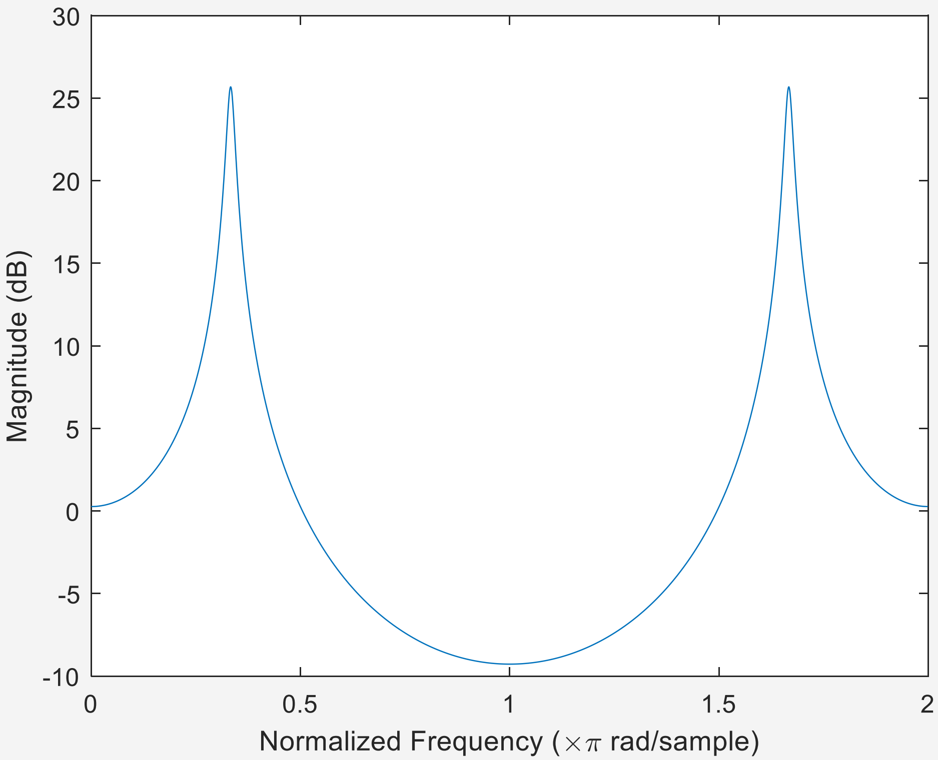
$$H(z) = \frac{z^2}{z^2 - 0.97z + 0.94}$$




```
%% Z transform
sym z;
Hz = 1/( 1 - 0.97*z^(-1) + 0.94*z^(-2));

[N,D] = numden(Hz);
pay = roots([0 0 100]);
payda = roots([94 -97 100]);

%% Z-plane
zplane(pay, payda);
```

MATLAB Code

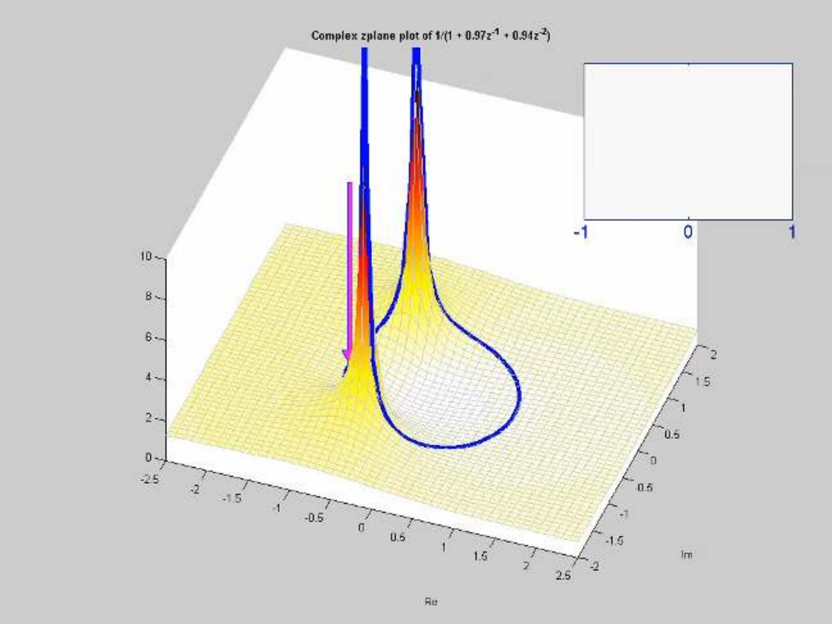
$$H(z) = \frac{z^2}{z^2 - 0.97z + 0.94}$$




```
%% Z transform
sym z;
Hz = 1/( 1 - 0.97*z^(-1) + 0.94*z^(-2));

[N,D] = numden(Hz);
pay = roots([0 0 100]);
payda = roots([94 -97 100]);

%% Z-plane
zplane(pay, payda);

%% Freq. Response
[h,w] = freqz([0 0 100],[94 -97
100],'whole',2001);
plot(w/pi,20*log10(abs(h)))
ax = gca;
ax.XTick = 0:.5:2;
xlabel('Normalized Frequency’)
ylabel('Magnitude (dB)')
```

MATLAB Code

$$H(z) = \frac{z^2}{z^2 - 0.97z + 0.94}$$




```
%% Z transform
syms a b;
Hz = 1/( 1 - 0.97*(a+1j*b)^(-1) + 0.94*(a+1j*b)^(-
2));

fsurf(abs(Hz));
xlim([-1 1]);
ylim([-1 1]);

hold on;
omega = 0:pi/100:2*pi;
s = exp(-1j*omega);
plot3(real(s),imag(s),2*ones(201,1),'LineWidth',3);
```

Exercise-1

EXERCISE 9.2: Determine the system function $H(z)$ of an FIR filter whose impulse response is

$$h[n] = \delta[n] - 7\delta[n-2] - 3\delta[n-3]$$

SOLUTION to EXERCISE 9.2: **DSP First 2e**

$$H(z) = \sum_{n=0}^{3} h[n]\, z^{-n}$$

EQUALS ONE AT n=0

$$= \sum_{n=0}^{3} \big(\delta[n] - 7\delta[n-2] - 3\delta[n-3] \big) z^{-n}$$

$$= z^{-0} - 7z^{-2} - 3z^{-3}$$

$$H(z) = 1 - 7z^{-2} - 3z^{-3}$$

McClellan, Schafer, and Yoder, *DSP First*, 2e, ISBN 0-13-065562-7

Exercise-2

EXERCISE 9.5: Use z-transforms to combine the following cascaded systems

$$w[n] = x[n] + x[n-1]$$
$$y[n] = w[n] - w[n-1] + w[n-2]$$

into a single difference equation for $y[n]$ in terms of $x[n]$.

SOLUTION to EXERCISE 9.5: **DSP First 2e**

$$w[n] = x[n] + x[n-1]$$
$$W(z) = X(z) + z^{-1}X(z)$$

DELAY PROPERTY

$$y[n] = w[n] - w[n-1] + w[n-2]$$
$$Y(z) = W(z) - z^{-1}W(z) + z^{-2}W(z)$$
$$Y(z) = (1 - z^{-1} + z^{-2})W(z)$$
$$= (1 - z^{-1} + z^{-2})(1 + z^{-1})X(z)$$
$$= (1 + z^{-3})X(z)$$

MULTIPLY POLYNOMIALS

$$Y(z) = X(z) + z^{-3}X(z)$$
$$y[n] = x[n] + x[n-3]$$

$w(z) = x(z) + x(z)\cdot z^{-1}$

$H_1(z) = 1 + z^{-1}$

$y(z) = w(z) - w(z)z^{-1} + w(z)\cdot z^{-2}$

$H_2(z) = 1 - z^{-1} + z^{-2}$

$H_1(z)\cdot H_2(z) = (1+z^{-1})(1-z^{-1}+z^{-2})$

$1 - \cancel{z^{-1}} + \cancel{z^{-2}} + \cancel{z^{-1}} - \cancel{z^{-2}} + z^{-3}$

$1 + z^{-3} = H(z)$

$H(z) = 1 + z^{-3} = \dfrac{Y(z)}{X(z)}$

$x[n] + x[n-3] = y[n]$

McClellan, Schafer, and Yoder, *DSP First, 2e*, ISBN 0-13-065562-7.

Exercise-3

Example 9-9: Consider a DTFT function expressed as

$$H(e^{j\hat{\omega}}) = (1 + \cos 2\hat{\omega})e^{-j3\hat{\omega}}$$

Using the inverse Euler formula for the cosine term gives

$$H(e^{j\hat{\omega}}) = \left(1 + \frac{e^{j2\hat{\omega}} + e^{-j2\hat{\omega}}}{2}\right) e^{-j3\hat{\omega}} = (e^{j\hat{\omega}})^{-3} + \tfrac{1}{2}(e^{j\hat{\omega}})^{-1} + \tfrac{1}{2}(e^{j\hat{\omega}})^{-5}$$

Making the substitution $e^{j\hat{\omega}} = z$ gives

$$H(z) = z^{-3} + \tfrac{1}{2}z^{-1} + \tfrac{1}{2}z^{-5} = \tfrac{1}{2}z^{-1} + z^{-3} + \tfrac{1}{2}z^{-5}$$

If the impulse response of the system is needed, the inverse z-transform gives

$$h[n] = \tfrac{1}{2}\delta[n-1] + \delta[n-3] + \tfrac{1}{2}\delta[n-5]$$

McClellan, Schafer, and Yoder, *DSP First, 2e*, ISBN 0-13-065562-7.
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